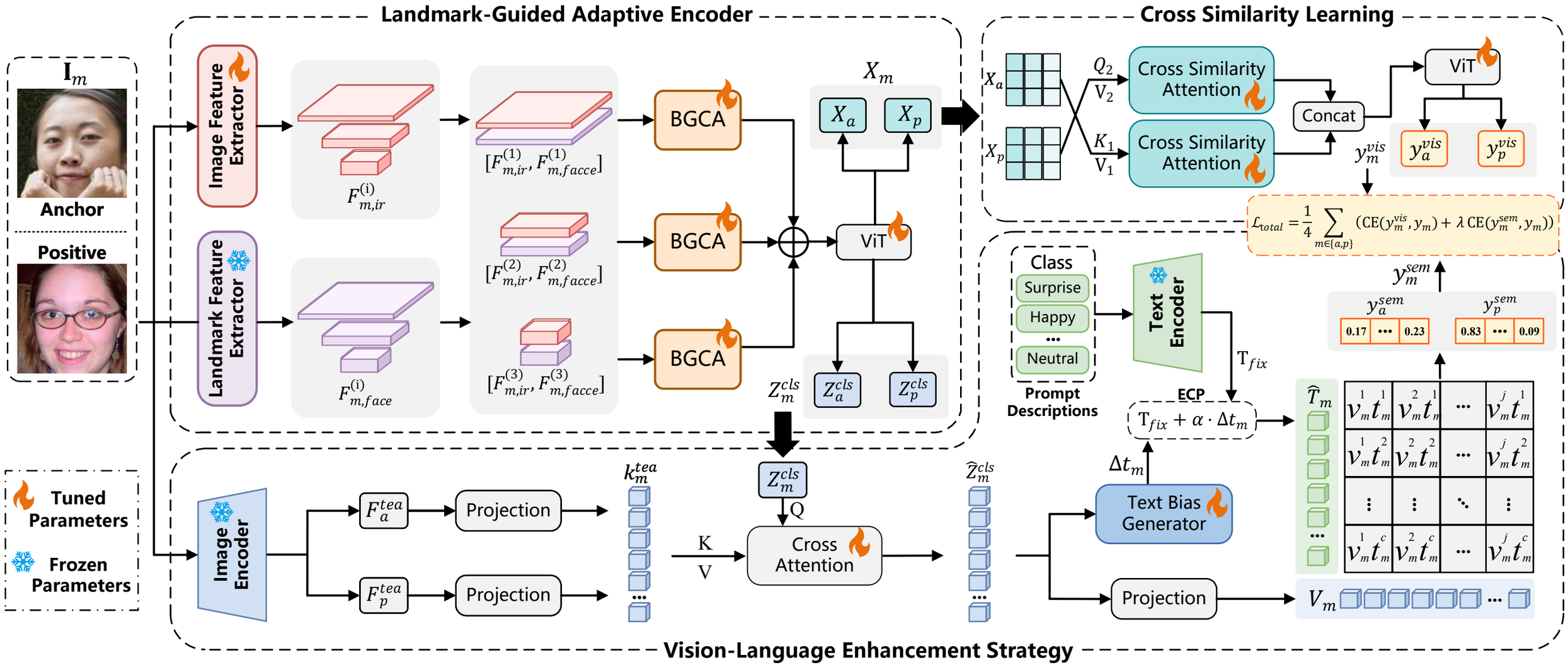
Landmark-Guided Adaptive Encoder
Cross Similarity Learning
Vision-Language Enhancement Strategy
I_m
Anchor
Positive
Image Feature Extractor
Landmark Feature Extractor
$F^{(i)}_{m,ir}$
$F^{(i)}_{m,face}$
$[F^{(1)}_{m,ir}, F^{(1)}_{m,facce}]$
$[F^{(2)}_{m,ir}, F^{(2)}_{m,facce}]$
$[F^{(3)}_{m,ir}, F^{(3)}_{m,facce}]$
BGCA
BGCA
BGCA
X_m
X_a
X_p
ViT
Z^{cls}_m
Z^{cls}_a
Z^{cls}_p
X_a
X_p
Q_2
V_2
K_1
V_1
Cross Similarity Attention
Cross Similarity Attention
Concat
ViT
y^{vis}_m
y^{vis}_a
y^{vis}_p
$\mathcal{L}_{total} = \frac{1}{4} \sum_{m \in \{a,p\}} (\mathrm{CE}(y^{vis}_m, y_m) + \lambda\, \mathrm{CE}(y^{sem}_m, y_m))$
Class
Surprise
Happy
...
Neutral
Prompt Descriptions
Text Encoder
T_{fix}
ECP
$T_{fix} + \alpha \cdot \Delta t_m$
Δt_m
Text Bias Generator
Projection
y^{sem}_m
y^{sem}_a
y^{sem}_p
0.17 ... 0.23
0.83 ... 0.09
$\hat{T}_m$
$v^1_m t^1_m$ $v^2_m t^1_m$ ... $v^j_m t^1_m$
$v^1_m t^2_m$ $v^2_m t^2_m$ ... $v^j_m t^2_m$
$v^1_m t^c_m$ $v^2_m t^c_m$ ... $v^j_m t^c_m$
V_m
Image Encoder
F^{tea}_a
F^{tea}_p
Projection
Projection
k^{tea}_m
K
V
Z^{cls}_m
Q
Cross Attention
$\hat{Z}^{cls}_m$
Tuned Parameters
Frozen Parameters